

Phạm Khánh Minh Mẫn

Backend Engineer · E-commerce API Integration · LLM-Agent Memory Research

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SUMMARY

Backend engineer with 5+ years in e-commerce, payments and multi-marketplace API integration, mostly in NestJS and TypeScript with event-driven work on AWS. Author of four open-source client libraries for the Shopee, TikTok Shop and Lazada Open APIs, published on npm.

Since 2026, a graduate student in Computer Science (research track) at Danang University of Science and Technology, interested in memory for LLM agents.

EXPERIENCE

DiproTech · Backend Developer

07/2025 - present

Backend for Japanese clients, often several projects at once. Client names withheld under contract.

Social App

12/2025 - 05/2026

Social network backend, team of 10. I built the backend and owned deployment.

- Set up the NestJS backend from an empty repo and built the CI/CD on GitHub Actions: dev and staging on Docker Compose behind Nginx (dev with auto-renewed TLS), production on AWS ECS.
- Production deploys on demand, per service or all at once, never two runs in parallel, and waits until ECS reports the service stable. A dev deploy that fails its health check rolls back to the previous image by itself.
- Tuned the PostgreSQL connection pool and indexes, made email unique only among active accounts so deleted users can sign up again, and built a shuffled feed that paginates stably without storing any order.

NestJS · TypeScript · PostgreSQL · Redis · Docker · GitHub Actions · AWS ECS / ECR · Nginx

Subscription billing

08/2026 - present

Rails billing for a publisher on Stripe and PayPal side by side, plus passkey sign-in on AWS Cognito.

- Made payment handling idempotent: each plan change is claimed by one conditional UPDATE, so Stripe and PayPal webhook retries never apply it twice.
- Moved receipt PDFs from wkhtmltopdf to headless Chrome, and wrote the row and the file upload in one transaction, so a failed upload retries cleanly with no orphaned data.
- Wrote the proration logic as plain Ruby with no database or gateway, unit-tested in RSpec; checked against the client's 28 official cases, it caught 14 errors in the spec itself.

Ruby on Rails 8 · PostgreSQL · Stripe · PayPal · Sidekiq · RSpec · AWS Cognito · Heroku

Factory planning system

08/2025 - present

Production scheduling, line simulation and live machine data, split across several Django services.

- Put schedule generation behind a Redis lock so two triggers can never build the same schedule at once, and moved heavy aggregation into Celery background jobs.
- Extended the Django models package that three services share as a versioned wheel, and carried its schema migrations through all of them.
- Made bulk updates atomic, integrated an external system over GraphQL, and worked on the Java OPC-UA client that reads machine data.

Python · Django · Celery · Redis · PostgreSQL · Docker · Java · OPC-UA · GraphQL

Exam-practice app

07/2025 - 10/2025

- Migrated the backend from PostgreSQL to Firestore, with batch import and export scripts for the question bank.
- Cut per-request work: stats update incrementally instead of recomputing a whole category, and Redis caches stats and exam history.

Python · FastAPI · Firestore · Firebase Auth · Redis

Devtify Technologies · Backend Developer

04/2022 - 06/2025

PaymentShield

01/2025 - 06/2025

Auto loan service between lenders, dealers and customers, team of 7.

- Started the codebase: NestJS, Docker, PostgreSQL, layered by responsibility.
- Webhooks publish to SQS and Lambda consumes them, so a slow downstream cannot block the callback.
- Failed jobs retry with backoff and land in a DLQ instead of disappearing.

NestJS · TypeScript · PostgreSQL · AWS Lambda · AWS SQS · BullMQ · Redis

ROUTINE

01/2024 - 03/2025

Orders, inventory and refunds unified across Shopee, Lazada and TikTok Shop, team of 10.

- One order model over three marketplaces, each with its own API shape and failure modes.
- Kept stock and finance in step with the NaviWorld ERP.
- Integrated three payment gateways and three shipping providers.

NestJS · TypeScript · PostgreSQL · Redis · Elasticsearch · BullMQ · Shopee API · TikTok Shop API · Lazada API

BEAUTYBOX / THEFACESHOP / REEBOK

04/2022 - 11/2023

Retail digital transformation for HSVGroup, team of 15.

- Designed the database and REST API.

NestJS · TypeScript · PostgreSQL · Redis · AWS EC2 · Docker

Vietnam People's Army · Military Service
Completed compulsory military service.

01/2020 - 01/2022

D-SOFT JSC · Backend Developer

03/2019 - 12/2019

SunWorld B2B Ticket

Ticket selection, online payment and e-ticket storage, team of 7.

- Designed and built the REST API on PHP / Laravel / SQL Server.

PHP · Laravel · SQL Server

SELECTED SYSTEMS

BattleCatsLab battlecatslab.fun

Personal project · public

Data and simulation site for a mobile game - 9,000 generated pages, a deterministic battle engine, and server-authoritative realtime PvP.

- Server owns the sim and ticks it at 30 Hz; clients send deploy intents and render snapshots, so a client cannot fake a result.
- Ranked ELO gate treats same-IP as telemetry only - shared Wi-Fi must not invalidate a real match; the abuse lockout derives from an append-only log, not a mutable counter.
- A golden-vector test pins fixed team pairs to their exact winner and frame count, so an engine change that would desync stored replays fails CI instead of corrupting them silently.

TypeScript · React · Vite · SQLite · Cloudflare Workers · Cloudflare R2 · SSE

OPEN-SOURCE PACKAGES

Shopee API Client (npm: shopee-api-client) - TypeScript client for Shopee Open API v2. Covers seller authorization, token management, orders, products, logistics, and payment escrow.

TikTok Shop API Client (npm: tiktokshops-api-client) - TypeScript API client for TikTok Shop Open API. Covers seller authorization, order APIs, product APIs, and fulfillment APIs.

Lazada API Client (npm: lazada-api-client) - TypeScript API client for Lazada Open API. Covers seller authorization, order APIs, and product APIs.

All-in-One Package (npm: shopee-tiktokshops-lazada-api) - Bundles the Shopee, TikTok Shop and Lazada SDKs into a single dependency, with versions kept in sync automatically.

SKILLS

Programming languages	TypeScript, JavaScript, Python, PHP, Ruby
Backend Frameworks	NestJS, Express, Laravel, Ruby on Rails, Django, FastAPI
Databases & Search	PostgreSQL, Redis, Elasticsearch, SQL Server, Firestore
Queues & Background Jobs	BullMQ, AWS SQS + DLQ, Sidekiq
Cloud & DevOps	AWS EC2 / S3 / ECS / ECR / Lambda / Cognito, Cloudflare Workers / R2 / D1, Docker, GitHub Actions, GitLab CI, Linux
Testing	RSpec, Vitest, pytest, Golden-vector regression tests
Payment Integrations	Stripe (Checkout, Subscriptions, Webhooks), PayPal Subscriptions, Fundiin, Payoo, ZaloPay, Recurring billing & proration, Webhook idempotency, Invoicing & consumption tax (適格請求書)
E-commerce Marketplaces	Shopee Open API v2, TikTok Shop API v2, Lazada Open Platform
Shipping & ERP	GHN, Ahamove, TikiNOW, NaviWorld ERP
Frontend	React, Next.js, Nuxt

EDUCATION

M.Sc. in Computer Science

2026 - present

Danang University of Science and Technology (DUT) - The University of Danang
Research track. Coursework in progress; thesis topic not yet decided.

B.Sc. in Information Technology

2015 - 2019

University of Science and Education - The University of Danang
Major in Information Technology.

RESEARCH

When Does Graph Memory Help Beyond Verbalized Relations? Isolating Representation from Structure in LLM-Agent Memory

2026 - present | Exploratory - no published results | Independent research - not committed as a thesis topic

Question. Graph memory is widely believed to help LLM agents remember better. But when graph memory is compared against flat memory, existing work changes TWO things at once: structure (the ability to traverse relations) and representation (information rewritten as typed relational sentences - unavoidable, because an LLM can only read text). So which of the two does the observed benefit belong to?

Method. Verbalize-control: three token-budget-matched conditions drawing from the same candidate edge pool and differing only in the selector - graph traversal, cosine truncated to the budget, and dump-everything. Plus a degree-preserving shuffled-edge control to separate “these particular edges” from “merely having a graph”. The design is pre-registered, evaluated on an external non-circular benchmark, and specifies the negative branch in advance (TOST equivalence testing).

Status, stated plainly: this is something I work on outside of client work, not a committed thesis topic. A 30-question feasibility probe has been run; the quality gate fired and returned “inconclusive” rather than a result. No publishable numbers.